

28) When the variable resistor is $0\ \Omega$, the pd across the $10\ \Omega$ resistor is $\frac{10}{20} \times 40V = 20V$.

When the variable resistor is $20\ \Omega$, the pd across the $10\ \Omega$ resistor is $\frac{10}{40} \times 40V = 10V$.

Answer A

$$E = 2.0 \times 10^5$$